

Week 10 - Radiosondes: Practice Questions

EART23001 Atmospheric Physics and Weather

Workload guide. The shaded question headings show the intended workload:

- **IN CLASS:** selected anchor problems that we may work through together in the timetabled session.
- **YOU SHOULD TRY:** expected independent practice after the session.
- **EXTRA CHALLENGE:** optional problems for students who want additional depth or a harder application.

Attempt the questions before looking at the worked answers. Show equations, substitutions and units for numerical work, and use sketches where they help. You are **not** expected to complete every Extra Challenge problem. A course-wide Symbols, Constants and Units sheet is provided separately; non-standard symbols are also defined locally.

The table below is a synthetic radiosonde profile designed for practice. Wind direction is the direction *from* which the wind blows. Treat the measurements as exact unless the question asks about uncertainty.

Height (m)	Pressure (hPa)	T ($^{\circ}\text{C}$)	RH (%)	Wind dir. ($^{\circ}$)	Speed (m s^{-1})
0	1010	14	75	220	4
500	955	10.5	82	230	6
1000	900	7	91	240	8
1500	845	3.5	97	250	10
2000	795	0	93	260	12
3000	700	-6	65	270	16
5000	540	-19	40	280	22
8000	360	-40	30	290	30
10000	265	-54	25	300	38
12000	195	-53	20	310	42

YOU SHOULD TRY Question 1: Which quantities are measured directly and which are derived?

Using the columns in the sounding table:

1. Identify the quantities measured by sensors carried on the radiosonde.
2. Explain how wind speed and direction are obtained during the flight.
3. Height is often calculated from pressure/GPS information rather than measured by a mechanical “height sensor”. State one reason why it is useful to report the profile versus both height and pressure.

YOU SHOULD TRY Question 2: Diagnose the thermodynamic structure of the sounding

Use the table to answer all parts.

1. Calculate the mean environmental lapse rate from the surface to 2.0 km.
2. Compare it with the DALR, 9.8 K km^{-1} , and state whether an unsaturated parcel is dry-statically stable through this layer.
3. Identify the layer most likely to contain cloud from the RH observations.
4. Estimate the freezing level.
5. Identify the main dry layer immediately above the likely cloud.
6. What temperature feature between 10 and 12 km suggests the tropopause has been reached?

YOU SHOULD TRY Question 3: Quantify the vertical wind shear

Between the surface and 10 km:

1. By how much does wind speed increase?
2. What is the mean increase in wind speed per kilometre over this depth?
3. By how many degrees does the reported wind direction change, and does it veer clockwise or back anticlockwise with height?
4. State one forecasting reason why this vertical change in wind cannot be inferred from surface observations alone.

EXTRA CHALLENGE Question 4: Why does the balloon expand and burst?

A weather balloon has volume 2.0 m^3 at launch, where $p_1 = 1010 \text{ hPa}$ and $T_1 = 287 \text{ K}$. Near the top of the profile, take $p_2 = 200 \text{ hPa}$ and $T_2 = 220 \text{ K}$. Neglect gas leakage and assume ideal-gas behaviour.

1. Calculate the balloon volume at the upper level using $pV/T = \text{constant}$.
2. By what factor has its volume increased?
3. If the envelope fails at about 8.0 m^3 , is a burst near this level plausible?

EXTRA CHALLENGE Question 5: Estimate ascent time and horizontal drift

Suppose the balloon ascends at a constant 5.0 m s^{-1} to 10 km. For a rough drift estimate, take the layer-mean horizontal wind speed to be 20 m s^{-1} .

1. How long does the ascent to 10 km take in seconds and minutes?
2. How far horizontally would the balloon travel at the assumed mean wind speed?
3. Give two reasons why the real trajectory would differ from this straight-line estimate.

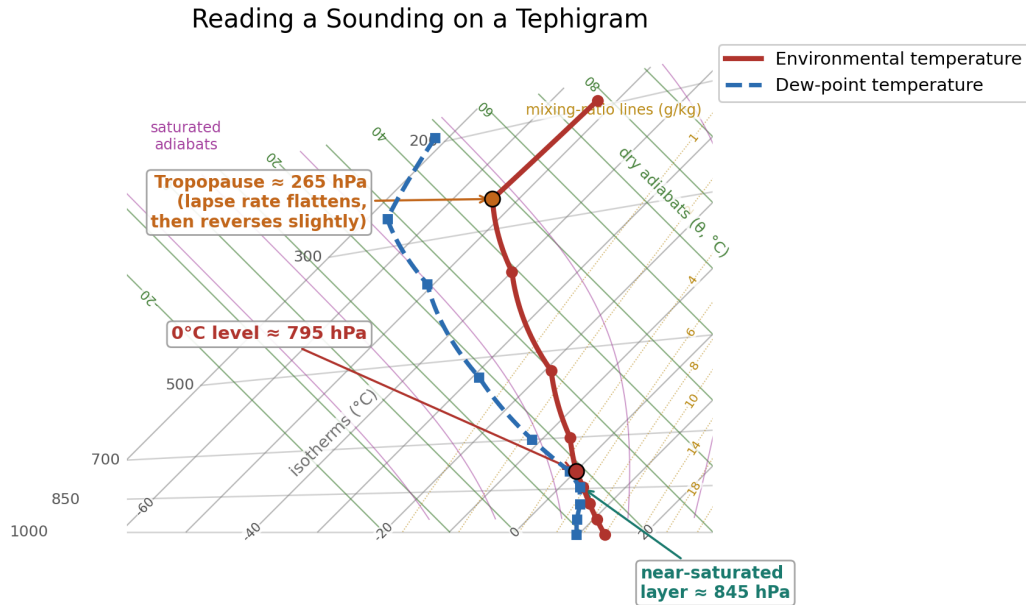
YOU SHOULD TRY Question 6: Use the sounding for an operational forecast

A precipitation system is approaching the launch site. Use only the sounding information to assess the following statements as *supported*, *not supported*, or *cannot be decided from the sounding alone*. Give a brief reason for each.

1. Low cloud is likely between about 1 and 2 km.
2. Any snow generated above 3 km must reach the surface as snow.
3. Aircraft could encounter a freezing level near 2 km.
4. Stronger winds occur aloft than at the surface.
5. The sounding proves that heavy rain will occur during the next hour.

IN CLASS Question 7: Read a plotted sounding rather than a table

The same synthetic sounding is plotted below on a simplified tephigram-style thermodynamic diagram. The solid purple curve is temperature and the dashed black curve is dew point. Grey guide lines show isotherms, dry adiabats and pressure levels. Use the diagram together with the table.

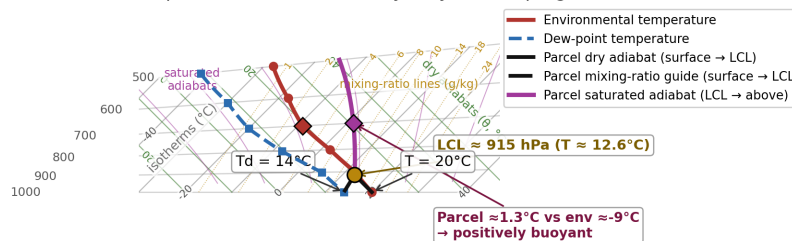


1. Identify the likely cloud layer from the separation between temperature and dew point.
2. Estimate the freezing level and tropopause height.
3. Where is the driest lower/mid-tropospheric air evident?
4. Explain one advantage of seeing the profile graphically rather than only as a table of numbers.

Worked example: finding the LCL and testing parcel buoyancy on a tephigram. Q7 used a tephigram-style diagram to read off an environmental profile. A tephigram can also be used to trace what happens to a *rising parcel*, which is the skill behind lifting-condensation-level and stability questions more generally.

A surface parcel has $T = 20^\circ\text{C}$ and $T_d = 14^\circ\text{C}$ at 1000 hPa (here, the same as the environment at the surface). The figure below draws two guide lines from this starting point: the parcel's *dry adiabat* (solid black), along which an unsaturated parcel cools at the dry adiabatic lapse rate, and its *mixing-ratio guide* (dash-dot black), along which the parcel's dew point falls as pressure drops, because its water-vapour mixing ratio r_v stays fixed until the parcel saturates.

Worked Example: LCL and Parcel Buoyancy on a Tephigram



Step 1 – find the LCL. The two guide lines cross at about 915 hPa, where the parcel's dry-adiabatic temperature ($\approx 12.6^\circ\text{C}$) exactly equals the dew-point value implied by its fixed mixing ratio. Below this pressure the parcel's dew point is below its temperature (unsaturated); above it, continuing along the dry adiabat would push the implied dew point above the temperature, which cannot happen – so this crossing is the Lifting Condensation Level (LCL), where the parcel first saturates and cloud base would form.

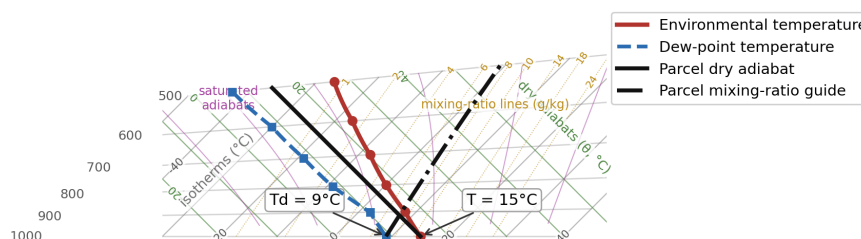
Step 2 – switch to a saturated adiabat. From the LCL, the parcel continues rising along a *saturated adiabat* (magenta) rather than a dry adiabat: condensation now releases latent heat into the parcel as it cools, so it cools more slowly than an unsaturated parcel would. Following this saturated adiabat up to 700 hPa gives a parcel temperature of about 1.3°C .

Step 3 – compare with the environment. The environmental temperature at 700 hPa here is about -9°C (red curve). The parcel ($\approx 1.3^\circ\text{C}$) is therefore several degrees warmer than its surroundings, so it is positively buoyant at 700 hPa and would continue to accelerate upward under its own buoyancy (ignoring virtual-temperature and entrainment effects) – the same reasoning used to identify genuinely unstable layers on a real sounding.

YOU SHOULD TRY Question 8: Practice: trace a parcel's ascent on a tephigram

The figure below shows an environmental sounding, together with a surface parcel of $T = 15^\circ\text{C}$ and $T_d = 9^\circ\text{C}$ at 1000 hPa. Its parcel dry adiabat (solid black) and mixing-ratio guide (dash-dot black) are drawn from the surface, exactly as in the worked example above.

Practice: Trace This Parcel's Ascent



1. By eye, estimate the pressure at which this parcel reaches its LCL.
2. From the LCL, the parcel continues along a saturated adiabat. Estimate its temperature at 700 hPa.

3. The environmental temperature at 700 hPa is -8°C . Is the parcel positively or negatively buoyant there?
4. Explain, in one or two sentences, why the parcel's path follows a different family of guide lines above the LCL than below it.